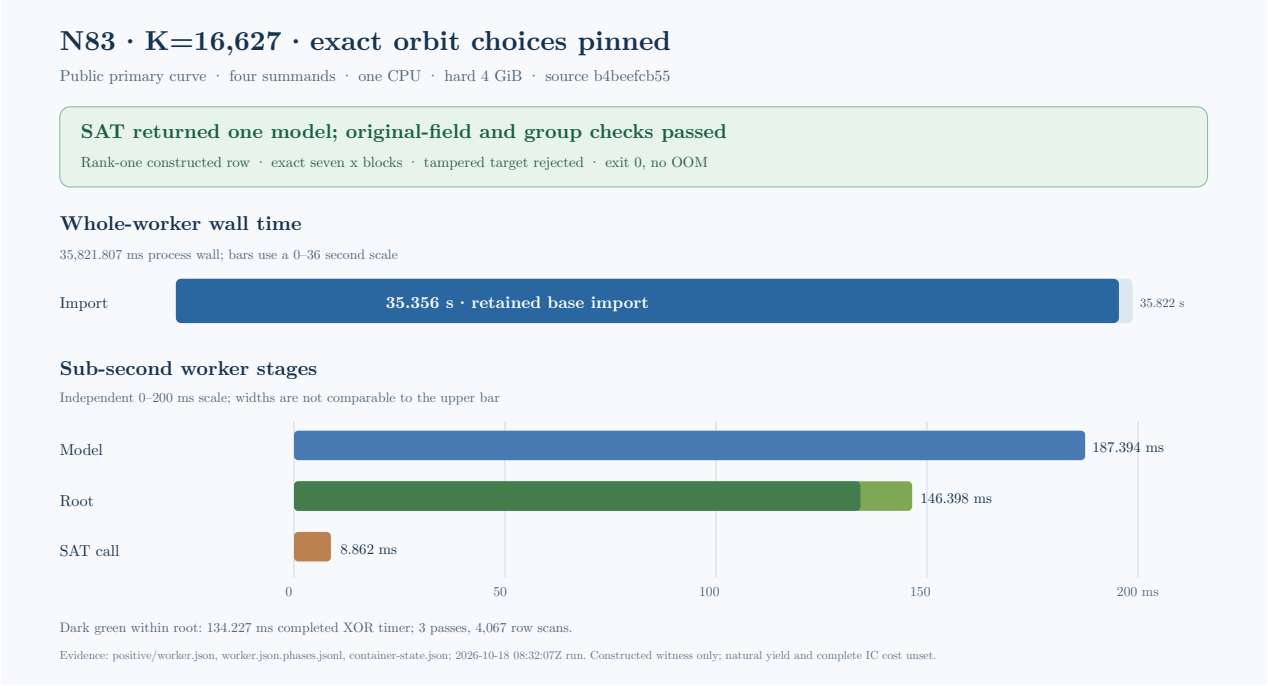
N83 K = 16,627: selector-pinned four-summand control

Run: 2026-10-18 08:32:07–08:32:43 UTC. **Worker result:** `PASS_SELECTOR_PINNED_WITNESS`. This is a constructed relation control on `icv1-f2m83-tm6151469093347-debefd74`. The exact prebuilt signed-Frobenius base has 16,627 columns and 2,760,082 stored points. The accepted source is commit `b4beefcb55f05713d1d63468fcbcd21e4339fa50`; `source-files.sha256`, `inputs-before-run.sha256`, `worker-binary.sha256` and `BUILD_RECEIPT.md` bind the executed worker and public inputs.



Frozen arm and observations

Measure	Recorded value
Resource envelope	one CPU; hard 4 GiB; zero swap; 1,800 s outer cap
Full model before choice pins	201,598 variables; 5,915,238 clauses; 3,901 XOR rows
Base import	35,356.152683 ms
Model construction	187.394292 ms
Root	`Fixpoint` in 146.397875 ms; all 201,598 variables assigned
Root XOR work	3 passes; 134,226,874 ns; 4,067 row scans
SAT call	`Sat` in 8.861792 ms; one model, zero conflicts
Worker wall	35,821.806933 ms
Container	exit 0; no OOM; 2,662,318,080-byte peak

Verification and scope

The worker checked all seven x blocks against the known public points, re-evaluated the original-field S3 chain, lifted the four summands to the stored base, matched the preflight row, obtained rank one and rejected an `R+G` target tamper. The raw worker, phase and progress records are in `positive/`, bound by `positive-files.sha256`. The small direct/observed controls passed; their timing spread is recorded in `VALIDATION.md`. For 20 AB/BA pairs of 100 checked solves per arm, the direct batch median was 8.842667 ms (8.514958–288.700333 ms range) and the observed batch median was 14.477375 ms (8.521667–115.515958 ms range). This shared-host spread is not used as an overhead correction or speed claim.

The separate native `replay/receipt.json` recomputed curve and subgroup membership, seven group x blocks, three original-field S3 equations and the phase-zero relation row from the compressed object. Changed-x and changed-row receipts were rejected. The worker did not export SAT assignment bits, so the separate replay does not check those bits. It also shares the library's field and curve arithmetic with the worker; it is a separate execution path, not independent arithmetic or second-host replication.

The earlier seven-x-block control reached root fixpoint with 63,910 propagated assignments, then reached its outer cap while searching. This choice-pinned arm reached root fixpoint with 201,598 assigned variables. The earlier arm used a different source revision and lacked the choice pins, so these wall values are not a paired speed ratio. This positive control does not measure natural relation yield, useful natural rank or complete cold IC cost. The next experiment separates selector and phase information under a common source. Canonical end-to-end cost and curve graphs remain unchanged.